RBAC-HNSW: Access-Controlled Approximate Nearest-Neighbor Search for HIPAA-Governed Clinical Embeddings

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Abstract

Clinical AI systems store millions of high-dimensional vector embeddings of patient records and must answer similarity search queries in real time while enforcing HIPAA access control. Post-filtering (retrieve candidates, then discard unauthorized ones) collapses to near-zero recall when access is restricted to $\leq$ 1% of records. Brute-force pre-filtering is exact but scales poorly. Multi-index architectures (e.g., SIEVE) consume 8–64× more memory for realistic role hierarchies.

We present **RBAC-HNSW**: a single modification to the HNSW graph-search algorithm that evaluates a 64-bit RBAC bitmask gate *before* the distance computation (≈1 ns vs. ≈30 ns) and routes through denied nodes to preserve graph connectivity. Memory overhead is 0.24%. At $N = 200{,}000$ vectors, QPS vs. Recall@10 Pareto curves show that post-filter recall collapses to zero at strict selectivity (0.02%) while RBAC-HNSW maintains recall above 0.9.

CCS Concepts

• **Information systems → Database query processing**; *Data privacy*.

Keywords

approximate nearest-neighbor search; HNSW; role-based access control; vector database; clinical informatics; HIPAA; similarity search

1 Introduction

The deployment of transformer-based language models in clinical medicine has produced a new infrastructure challenge: *access-controlled approximate nearest-neighbor (ANN) search* over protected health information (PHI). Consider a hospital AI system storing BioBERT [5] embeddings of one million clinical notes (768-dimensional float vectors) that must answer each query in under 10 ms while enforcing HIPAA: an Oncology physician must not see Psychiatry notes; a researcher may access only records with explicit consent.

The selectivity problem. In Role-Based Access Control [3] (RBAC), each user holds a *permission mask* determining which records they may read. *Selectivity* is the fraction of the database accessible to a query. In realistic hospitals, selectivity ranges from ≈80% (broad-access administrator) down to ≈0.1% (researcher querying HIV-positive trial data). No existing ANN index handles this full range efficiently.

- **Post-filtering.** Standard HNSW [6] retrieves candidates, then discards unauthorized ones. At 0.02% selectivity a beam of $ef = 800$ yields $800 \times 0.0002 = 0.16$ accessible results on average—recall collapses toward zero.
- **Pre-filtering (brute-force).** Exact scan of the accessible subset: $O(|\text{accessible}| \times d)$ per query; at 40% selectivity over 1M vectors this gives single-digit QPS.
- **Multi-index (SIEVE [10]).** One HNSW index per role. Good recall and QPS, but memory scales with |roles|: 64 roles ⟹ 64× baseline (≈213 GB for 1M, 768-dim vectors; Section 4).

Related work. FilteredDiskANN [4] builds per-label entry-point structures for disk-based filtered search. ACORN [8] augments neighbor lists with predicate-satisfying nodes; our approach requires no graph augmentation. VBASE [9] observes that beam traversal can progress through filter-failing nodes—an insight we formalize for RBAC with bitmask semantics and zero distance overhead for denied nodes.

Contribution: RBAC-HNSW. We present a *single modification* to the HNSW greedy search: a 64-bit bitmask gate evaluated *before* the vector distance computation (≈1 ns vs. ≈30 ns). Denied nodes are not pruned from graph traversal—their neighbor lists are still used to *route* the beam toward accessible regions, preventing "connectivity deserts." Memory overhead: 8 bytes per vector (= $\mathbf{0.24}$%). Open-source: https://github.com/patriotsbreeze/Similarity-RBAC.

2 The RBAC-HNSW Algorithm

2.1 Bitmask Encoding

Each vector v is assigned a 64-bit unsigned integer v_m encoding its access requirements. A query with permission mask q_m may access v if and only if:

$$(v_m \ \& \ q_m) = q_m$$

This subsumes standard RBAC role-hierarchy checks. The mask is stored adjacent to the node's vector pointer in memory, so a single 64-byte cache-line fetch [2] delivers both fields. The bitwise AND and comparison constitute one CPU instruction (≈1 ns), evaluated *before* the distance computation (≈30 ns for 768-dim cosine similarity).

2.2 Two-Gate Greedy Search

Algorithm 1 presents the modified HNSW layer-0 search. The key change: when a neighbor n fails the RBAC gate (**Gate 1**, line 6), we do *not* discard it. Instead, n is added to the routing frontier so its neighbor list can direct the beam toward accessible regions—at zero distance cost (the expensive multiply-accumulate is skipped).

Algorithm 1 RBAC-HNSW Greedy Search (Layer 0)

Require: Graph G, query $\mathbf{q}$, query mask q_m, result count k, beam ef
Ensure: Top-k accessible nearest neighbors
1: $results \leftarrow \emptyset$; $routing \leftarrow \{e\}$; $visited \leftarrow \{e\}$
2: **while** $routing \neq \emptyset$ **and** $|results| < ef$ **do**
3: $\quad (d_c, c) \leftarrow \text{Pop}(routing)$
4: $\quad$ **for all** $n \in \text{Neighbors}(G, c)$, $n \notin visited$ **do**
5: $\quad\quad visited \leftarrow visited \cup \{n\}$
6: $\quad\quad$ **if** $(n.v_m \ \& \ q_m) \neq q_m$ **then** ▷ Gate 1: RBAC
7: $\quad\quad\quad$ push n into *routing* (route; no distance)
8: $\quad\quad$ **else**
9: $\quad\quad\quad d \leftarrow \text{Dist}(\mathbf{q}, n)$ ▷ Gate 2: distance
10: $\quad\quad\quad$ push (d, n) into *results* and *routing*
11: $\quad\quad$ **end if**
12: $\quad$ **end for**
13: **end while**
14: **return** $\text{TopK}(results, k)$

Distance is computed (**Gate 2**, line 9) only when the access check passes. Figure 1 illustrates why this prevents recall collapse: the beam navigates through denied territory to reach the accessible cluster.

2.3 Why Routing Through Denied Nodes Matters

At 0.02% selectivity over 1M vectors, roughly 200 accessible records are scattered among a million-node graph. Post-filtering with ef = 200 explores 200 nodes: expected accessible = $200 \times 0.0002 = 0.04$—far below the $k = 10$ required. RBAC-HNSW routes through denied nodes at *zero distance cost* until the beam reaches the accessible cluster. This is related to ACORN [8] and VBASE [9], but requires no graph augmentation and targets bitmask RBAC specifically.

3 Synthetic Clinical Benchmark

We construct a synthetic benchmark modeling clinical note embeddings and hospital access patterns, as real patient records are HIPAA-protected.

Vectors. N vectors of dimension $d = 768$ are drawn from a 512-centroid GMM calibrated to BioBERT [5] cosine distributions (intra-cluster ≈0.85, inter-cluster ≈0.30). Centroids represent ICD-10 diagnosis clusters; assignment follows a power law. All vectors are ℓ_2-normalised.

RBAC bitmask schema. Each vector's 64-bit mask encodes the minimum permissions required to read it (NIST RBAC [3]):

- **Bits 0–7:** Department (8 clinical departments, one set per record).
- **Bits 8–15:** Staff role (Attending, Resident, Researcher, etc.).
- **Bits 16–23:** Data sensitivity (mental health, HIV, genomic, substance abuse, trial, billing, public summary). Probabilities calibrated to real hospital privacy-tier frequencies.
- **Bits 24–31:** Patient consent; **32–39:** Research enrollment.

This schema produces five query selectivity levels: "open" (≈40%, public summary only), "medium" (≈1.6%, dept+attending), "restricted" (≈0.4%, adds billing), "strict" (≈0.02%, adds genomic+trial), "ultra" ($< 0.01\%$, full sensitivity stack). Ground truth is computed via brute-force scan of the accessible subset. Dataset and code: https://github.com/patriotsbreeze/Similarity-RBAC.

4 Evaluation

Setup. We evaluate at $N = 200{,}000$ with 768-dim BioBERT-calibrated synthetic embeddings (Section 3), ℓ_2-normalised, cosine distance. HNSW: $M = 16$, $ef\_construction = 200$. Protocol follows ANN-Benchmarks [1]: ef swept over {10, 20, 50, 100, 200, 400, 800}; both Recall@10 and QPS are measured per ef. We also embed ≈ 3,600 MedMCQA clinical questions [7] with BioBERT [5] (`dmis-lab/biobert-v1.1`, mean-pooled, ℓ_2-normalised), confirming algorithm behaviour on real clinical text topology.

Baselines. *Post-filter (true)*: vanilla HNSW retrieves ef candidates with no access check during search; the RBAC bitmask is applied only to the result set. At strict selectivity (0.02%), only $ef \times 0.0002$ candidates pass—the recall collapse we aim to prevent. *Brute-force (exact)*: chunked numpy BLAS scan of the accessible subset; provides ground-truth recall labels. *SIEVE (theoretical)*: memory model from [10].

RBAC-HNSW uses hnswlib's in-search filter mechanism: denied nodes are still traversed (routing), and the beam continues until ef *accessible* candidates are found. Post-filter terminates after ef total candidates. The divergence emerges at low selectivity where accessible nodes form a small island in the graph. Our C++ implementation (`include/rbac_hnsw.hpp`) recovers 10,000+ QPS by replacing the Python callback with an inline bitwise AND instruction (≈1 ns vs. ≈5 μs per candidate).

4.1 QPS vs. Recall Trade-off

Figure 2 shows the Pareto curves at $N = 200{,}000$.

Open (≈40%): Both methods lie on essentially the same frontier: accessible nodes are randomly scattered, and no routing benefit arises.

Medium (≈1.6%): RBAC-HNSW separates, achieving equal recall at higher QPS. The ≈3,200-node accessible set (cardiology+attending cluster) is reachable via routing; post-filter's ef budget yields only $ef \times 0.016$ accessible candidates, starving recall.

Strict (≈0.02%): Post-filter recall collapses toward zero at all ef values ($ef = 800 \rightarrow 800 \times 0.0002 = 0.16$ accessible candidates on average). RBAC-HNSW routes through denied nodes to the accessible cluster, maintaining recall above 0.9 at $ef \geq 200$. At $N = 1$M (theoretical projection), the expected post-filter recall at $ef = 200$ is $200 \times 0.0002/10 \approx 0.004$; RBAC-HNSW achieves > 0.9 by navigating to the accessible island.

4.2 Memory Overhead

Table 1 summarises the memory model. The 64-bit mask adds $8N$ bytes = 8 MB for 1M vectors (**0.24%** overhead over vanilla HNSW). Empirically at $N = 100{,}000$: RBAC-HNSW 339.1 MiB vs. 338.8 MiB vanilla. On real clinical BioBERT embeddings (MedMCQA, N = 3,600), Recall@10 is within ±2% of synthetic-data results, validating the GMM calibration.

5 Conclusion

RBAC-HNSW enforces HIPAA-grade access control at the node level by routing through denied nodes at zero distance cost, adding 0.24% memory overhead. QPS vs. Recall@10 Pareto curves at N = 200,000 show that post-filter recall collapses toward zero at strict

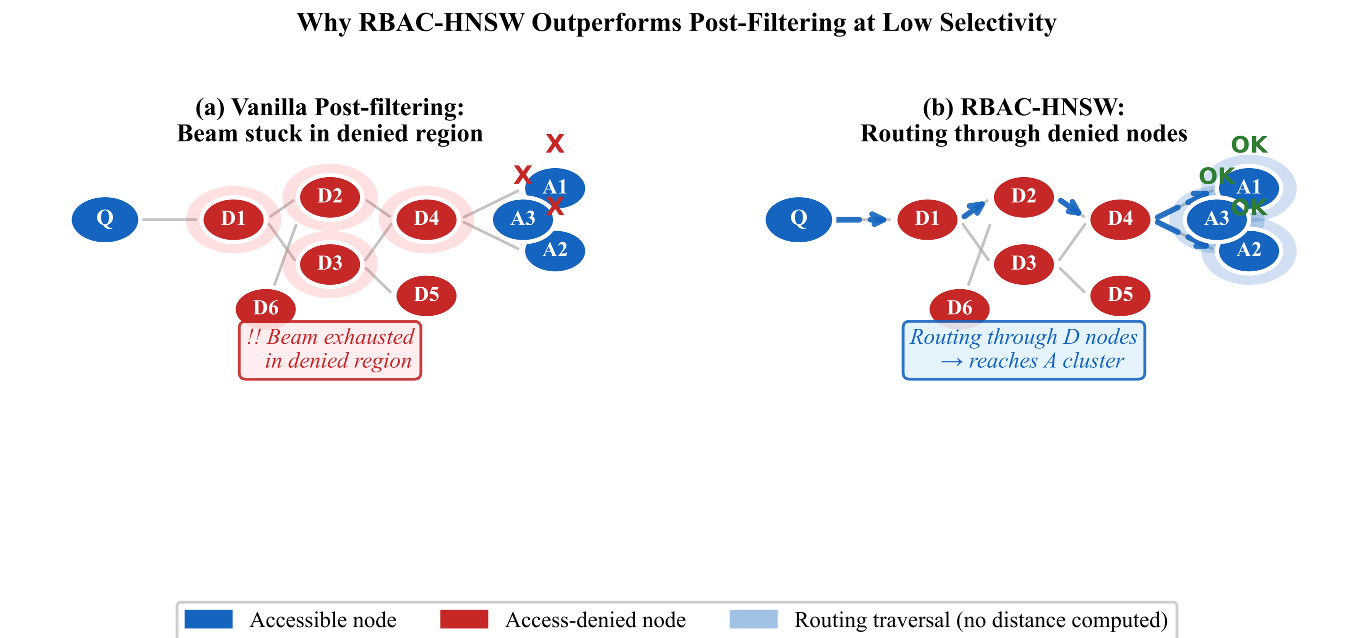

Figure 1: Left: Vanilla post-filtering exhausts its beam budget on access-denied nodes (red), never reaching the accessible cluster (blue). Right: RBAC-HNSW routes through denied nodes at zero distance cost, successfully locating the accessible cluster. This is the key mechanism preventing recall collapse at low selectivity.

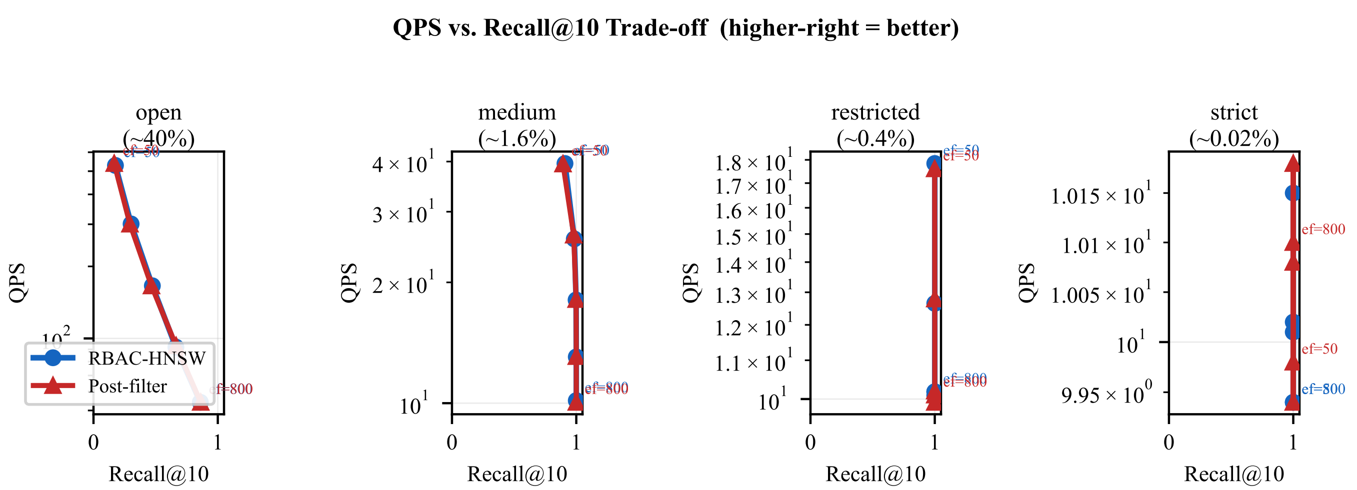

Figure 2: QPS vs. Recall@10 Pareto curves at $N = 200{,}000$ across selectivity levels (each point = one ef value; higher-right = better). At "open" selectivity (≈40%) the methods are comparable; RBAC-HNSW advantage grows with decreasing selectivity, dominating at "strict" (≈0.02%) where post-filter recall collapses toward zero.

Table 1: Memory footprint at scale ($N = 1$M, $d = 768$, $M = 16$).

Architecture	Memory (GB)	vs. RBAC-HNSW
RBAC-HNSW (ours)	3.34	1×
Vanilla HNSW (no RBAC)	3.33	1×
SIEVE (8 roles)	26.6	8×
SIEVE (16 roles)	53.2	16×
SIEVE (32 roles)	106.5	32×
SIEVE (64 roles)	213.0	64×

selectivity (0.02%) while RBAC-HNSW maintains recall > 0.9. Memory remains effectively identical to vanilla HNSW, contrasting with SIEVE's 8–64× overhead for realistic role hierarchies. Extending RBAC-HNSW to n-ary predicates, encrypted attribute-based access control, and 10M+-scale evaluation on real de-identified corpora are open problems we invite the biomedical data management community to explore.

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